Lecture 2 Time evolution operators in quantum statistical mechanics

1. Shrödinger eq.

$$H_t \psi_t = i\hbar \frac{\partial \psi_t}{\partial t} \equiv i\hbar \partial_t \psi_t \qquad (1)$$

- Introduce $U(t,t')$: $\boxed{\psi_t = U(t,t')\, \psi_{t'}} \rightarrow \boxed{U(t,t) = 1} \qquad (2)$

- Eq: $\boxed{U(t,t')U(t',t'') = U(t,t'')} \qquad (3)$

$$i\hbar \partial_t \psi_t = i\hbar \left(\partial_t U(t,t')\right)\psi_{t'} \equiv H_t \psi_t = H_t U(t,t')\, \psi_{t'}$$

$$\hookrightarrow \boxed{i\hbar \partial_t U(t,t') = H_t U(t,t')} \qquad (4)$$

- Differentiate wrt t': $\psi_t = U(t,t')\,\psi_{t'}$

$$\hookrightarrow 0 = \left(\partial_{t'} U(t,t')\right)\psi_{t'} + U(t,t')\,\partial_{t'}\psi_{t'} \qquad (5)$$

$$= \left(\partial_{t'} U(t,t')\right)\psi_{t'} + U(t,t')\frac{1}{i\hbar} H_{t'}\psi_{t'} \rightarrow \boxed{i\hbar \partial_{t'} U(t,t') = -U(t,t') H_{t'}}$$

- **Unitarity**

$$\hbar i \partial_t U(t,t') = H_t U(t,t') \leftarrow \text{take } \dagger \text{ of both sides}$$

unitary

$$-\hbar i \partial_t U^\dagger(t,t') = U^\dagger(t,t') H_t$$

Also, consider

$$i\hbar \partial_t \left(U^\dagger(t,t') U(t,t')\right) = i\hbar\left[\left(\partial_t U^\dagger(t,t')\right)U(t,t') + U^\dagger(t,t')\left(\partial_t U(t,t')\right)\right]$$

$$= -U^\dagger H_t U + U^\dagger H_t U = 0 \Rightarrow U^\dagger(t,t')U(t,t') = \text{Constant.}$$

At $t=t' \rightarrow U^\dagger(t,t)U(t,t) = 1 \Rightarrow \text{Const} = 1 \Rightarrow \boxed{U^\dagger(t,t')U(t,t') = 1}$

Therefore, if we define the inverse operator via

$$U^{-1}(t,t')\,U(t',t) = 1 \Rightarrow U^{-1}(t,t') = U^\dagger(t',t)$$

Also, $1 = U(t',t)\,U(t,t') \Rightarrow \boxed{U^\dagger(t,t') = U(t',t) \equiv U^{-1}(t',t)} \qquad (6)$

2. Time ordering

$$i\hbar\, \partial_t\, \mathcal{U}(t,t') = H_t\, \mathcal{U}(t,t')$$

Integrate between t' and t $(t' < t)$ wrt t:

$$-\underbrace{\mathcal{U}(t',t')}_{=1} + \mathcal{U}(t,t') = \frac{1}{i\hbar}\int_{t'}^{t} dt_1\, H_{t_1}\, \mathcal{U}(t_1,t')$$

$$\hookrightarrow \mathcal{U}(t,t') = 1 + \frac{1}{i\hbar}\int_{t'}^{t} dt_1\, H_{t_1}\,\mathcal{U}(t_1,t') = 1 - \frac{i}{\hbar}\int_{t'}^{t} dt_1\, H_{t_1}\,\mathcal{U}(t_1,t')$$

Iterate:

$$\mathcal{U}(t,t') = 1 + \sum_{n=1}^{\infty} (-i/\hbar)^n \int_{t'}^{t} dt_1 \int_{t'}^{t_1} dt_2 \cdots \int_{t'}^{t_{n-1}} dt_n\, \underbrace{H_{t_1} H_{t_2} H_{t_3} \cdots H_{t_n}} \qquad (7)$$

times in descending → order
or in ascending order ←

$$t_1 > t_2 > \dots > t_n$$

This can be written via $\hat{T}$ which orders operators in the correct order (← ascending):

$$\mathcal{U}(t,t') = 1 + \sum_{n=1}^{\infty} \frac{(-i/\hbar)^n}{n!} \int_{t'}^{t} dt_1 \int_{t'}^{t} dt_2 \cdots \int_{t'}^{t} dt_n\, \hat{T}\left[H_{t_1} H_{t_2} \cdots H_{t_n}\right]$$

$$\hookrightarrow \boxed{\mathcal{U}(t,t') \equiv \hat{T} \exp\left[-\frac{i}{\hbar}\int_{t'}^{t} d\tau\, H_\tau\right]} \qquad \boxed{t > t'} \qquad (8)$$

$\boxed{\text{For } \hat{H} \text{ not depending on time, } \mathcal{U}(t,t') = e^{-iH(t-t')/\hbar}}$

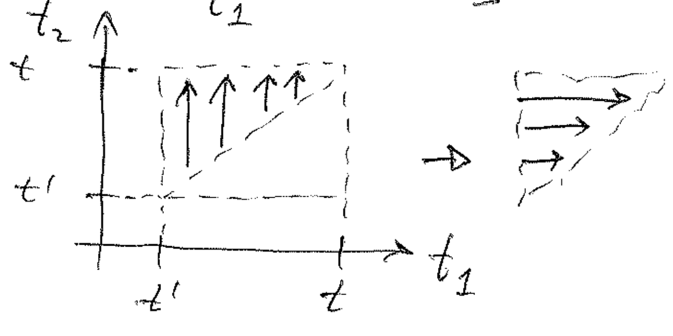
Proof

(i) $n=2$

$$\int_{t'}^{t} dt_1 \int_{t'}^{t} dt_2\, T(H_{t_1} H_{t_2}) = \int_{t'}^{t} dt_1 \left[\int_{t'}^{t_1} dt_2\, H_{t_1} H_{t_2} + \int_{t_1}^{t} dt_2\, H_{t_2} H_{t_1}\right]$$

The 2nd = 1st. Indeed,

$$\int_{t'}^{t} dt_1 \int_{t_1}^{t} dt_2\, H_{t_2} H_{t_1} =$$



$$= \int_{t'}^{t} dt_2 \int_{t'}^{t_2} dt_1\, H_{t_2} H_{t_1} = |t_2 \leftrightarrow t_1| = \int_{t'}^{t} dt_1 \int_{t'}^{t_1} dt_2\, H_{t_1} H_{t_2} \equiv \text{1st term}$$

↳ for $n=2$ we get $1/2!$ disappearing and the integrals with correct limits.

(ii) Let us now differentiate both sides of the solution for $U(t,t')$, Eq. (8):

$$i\hbar\,\partial_t U(t,t') = i\hbar\,\partial_t \left[1 + \left(-\frac{i}{\hbar}\right)\int_{t'}^{t} dt_1 H_{t_1} + \sum_{n=2}^{\infty} \frac{(-i/\hbar)^n}{n!} \int_{t'}^{t} dt_1 \ldots \hat{T}(H_{t_1}\ldots)\right]$$

$$= H_t + \sum_{n=2}^{\infty} i\hbar \frac{(-i/\hbar)^n}{n!} \left\{ \int_{t'}^{t} dt_2 \ldots \hat{T}(H_t H_{t_2}\ldots) + \int_{t'}^{t} dt_1 \int_{t'}^{t} dt_3 \ldots \hat{T}(H_{t_1} H_t H_{t_2}\ldots)\right.$$

$$\left. + \ldots + \int_{t'}^{t} dt_1 \ldots \int_{t'}^{t} dt_{n-1}\, \hat{T}\left(H_{t_1} H_{t_2} \ldots H_{t_{n-1}} H_t\right)\right\}$$

Since $t >$ any of the times $t_1, \ldots, t_n$, H_t can be taken to the left from $\hat{T}(\ldots)$:

$$i\hbar\,\partial_t U(t,t') = H_t + H_t \sum_{n=2}^{\infty} i\hbar \frac{(-i/\hbar)^n}{n!} \left\{ \int_{t'}^{t} dt_2 \int_{t'}^{t} dt_3 \ldots \hat{T}(H_{t_2} \ldots H_{t_n}) + \right.$$

$$\left. + (\text{similar terms, which after changing variables become identical}) \right\}$$

$$= H_t \left[1 + \sum_{n=2}^{\infty} i\hbar \frac{(-i/\hbar)^{n-1}}{(n-1)!} \left(-\frac{i}{\hbar}\right) \int_{t'}^{t} dt_2 \ldots \int_{t'}^{t} dt_n\, \hat{T}(H_{t_2} \ldots H_{t_n}) \right]$$

$$= H_t \left[1 + \sum_{m=1}^{\infty} \frac{(-i/\hbar)^m}{m!} \int_{t'}^{t} dt_1 \ldots \int_{t'}^{t} dt_m\, \hat{T}(H_{t_1} \ldots H_{t_m}) \right] \equiv H_t U(t,t')$$

which is exactly (8).

antichonological order

• Similarly,
$$\boxed{U(t,t') = \tilde{T} \exp\left[+\frac{i}{\hbar} \int_{t}^{t'} d\tau\, H_\tau\right], \quad t' > t} \qquad (9)$$

3. Quantum statistical mechanics

$\boxed{\mathcal{H} = H - \mu N}$!!!
everywhere here

- In equilibrium $\hat{\rho} = \frac{1}{Z} e^{-\beta(\hat{H} - \mu\hat{N})}, \quad Z = \mathrm{Tr}(e^{-\beta\mathcal{H}})$ (10)

Average: $\langle A \rangle = \mathrm{Tr}(\hat{A}\hat{\rho})$ (11)

- In nonequilibrium $\hat{\rho}_t$ satisfies Liouville equation:

$$\boxed{i\hbar\partial_t \rho_t = [\mathcal{H}_t, \rho_t]} \tag{12}$$

↳ may depend on t explicitly

- ~~This~~ If $\mathcal{H}$ does not depend explicitly on time,

$$\boxed{\hat{\rho}_t = e^{-i\mathcal{H}(t-t_0)/\hbar}\, \hat{\rho}_{t_0}\, e^{i\mathcal{H}(t-t_0)/\hbar}} \tag{13}$$

Indeed,

$$i\hbar\partial_t \hat{\rho}_t = i\hbar\, e^{-i\mathcal{H}(t-t_0)/\hbar}\left(-\frac{i\mathcal{H}}{\hbar}\rho_{t_0} + \rho_{t_0}\frac{i\mathcal{H}}{\hbar}\right)e^{i\mathcal{H}(t-t_0)/\hbar}$$

$$= \mathcal{H}\rho_t - \rho_t\mathcal{H} = [\mathcal{H}, \rho_t] \quad \blacktriangleleft$$

- In general, when $\mathcal{H} = \mathcal{H}_t$, the solution is:

$$\boxed{\rho_t = U(t,t_0)\, \rho_{t_0}\, U^\dagger(t,t_0)} \tag{14}$$

Indeed,

$$i\hbar\partial_t \rho_t = \left(i\hbar\partial_t U(t,t_0)\right)\rho_{t_0}U^\dagger + U\rho_{t_0}\left(-i\hbar\partial_t U(t,t_0)\right)^\dagger$$

$$= \mathcal{H}_t \underbrace{U(t,t_0)\rho_{t_0}U^\dagger}_{\rho_t} - U\rho_{t_0}\left(\mathcal{H}_t U(t,t_0)\right)^\dagger$$

$$= \mathcal{H}_t\rho_t - \overbrace{U\rho_{t_0}U^\dagger}^{\rho_t}\mathcal{H}_t = [\mathcal{H}_t, \rho_t] \quad \blacktriangleleft$$

- Assume that at t_0 the system was at equilibrium:

$$\rho_{t_0} = \frac{1}{Z}e^{-\beta\mathcal{H}_0}, \quad \mathcal{H}_0 = \mathcal{H}_{t_0}, \quad Z = \mathrm{Tr}(e^{-\beta\mathcal{H}_0}) \tag{15}$$

• The average of $\hat{O}$ at time t:

$$\langle \hat{O} \rangle_t = \mathrm{Tr}(\hat{\rho}_t \hat{O}) = \mathrm{Tr}\left(\mathcal{U}(t,t_0)\, \rho_{t_0}\, \mathcal{U}^\dagger(t,t_0)\, \hat{O} \right)$$

$$= \mathrm{Tr}\left(\rho_{t_0} \underbrace{\mathcal{U}^\dagger(t,t_0)\, \hat{O}\, \mathcal{U}(t,t_0)}\right) \equiv \mathrm{Tr}\left(\rho_{t_0} \underbrace{\hat{O}_H(t)} \right) \qquad (16)$$

Heisenberg representation of $\hat{O}$

We slightly rewrite it:

$\mathcal{U}(t,t') = e^{-i(t-t')\mathcal{H}/\hbar}$ for constant $\mathcal{H}$. Then, take $\mathcal{H}_0 = \mathcal{H}_{t_0}$:

$$\mathcal{U}(t_0 - i\beta\hbar, t_0) = e^{-\beta \mathcal{H}_0}$$

since $\mathcal{U}^\dagger(t,t_0) = \mathcal{U}(t_0,t)$

$$\hookrightarrow \boxed{\langle \hat{O} \rangle_t = \frac{\mathrm{Tr}\left[\mathcal{U}_0(t_0 - i\beta\hbar, t_0)\, \mathcal{U}(t_0,t)\, \hat{O}\, \mathcal{U}(t,t_0) \right]}{\mathrm{Tr}\left[\mathcal{U}_0(t_0 - i\beta\hbar, t_0) \right]}} \qquad (17)$$

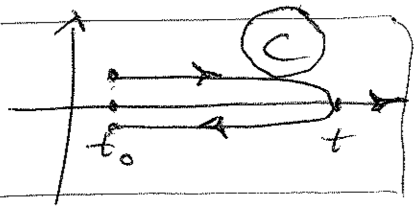
where $\mathcal{U}_0(t,t')$ is the evolution operator based on $\mathcal{H}_0$.

• We shall now prove that

$$\boxed{\mathcal{U}(t_0,t)\, \hat{O}\, \mathcal{U}(t,t_0) = \left(\tilde{T} \exp\left[+\frac{i}{\hbar} \int_{t_0}^{t} \mathcal{H}_\tau \, d\tau \right] \right) \hat{O} \left(\hat{T} \exp\left[-\frac{i}{\hbar} \int_{t_0}^{t} \mathcal{H}_\tau \, d\tau \right] \right)} \qquad (18)$$

can also be written as:

$$\boxed{\hat{T}_c \left(\exp\left[-\frac{i}{\hbar} \int_C \mathcal{H}_\tau \, d\tau \right] \cdot \hat{O} \right) \quad \text{with}} \qquad (19)$$



ordering on the countour

<u>Proof</u>

In (18) the order of operators, from right to left, is:

$$\underset{t_0 \leftarrow t}{\mathcal{H}_{\tau_n'} \dots \mathcal{H}_{\tau_2'} \mathcal{H}_{\tau_1'}} \hat{O} \underset{t \;\; \leftarrow \;\; t_0}{\mathcal{H}_{\tau_n} \dots \mathcal{H}_{\tau_2} \mathcal{H}_{\tau_1}} \rightarrow \mathcal{H}_{\tau_n'} \dots \mathcal{H}_{\tau_1'}\, \hat{O}_t\, \mathcal{H}_{\tau_n} \dots \mathcal{H}_{\tau_1}$$

So it is convenient to think that $\hat{O}$ acts at time t, $\hat{O} \Rightarrow \hat{O}_t$ in the following [it may actually be explicitly time dependent!]. Then:

$$\hat{T}_c \left(\exp\left[-\frac{i}{\hbar} \int_C \mathcal{H}_\tau \, d\tau \right] \cdot \hat{O}_t \right) = \sum_{n=0}^{\infty} \left(-\frac{i}{\hbar} \right)^n \frac{1}{n!} \int_C dt_1 \int_C dt_2 \dots \int_C dt_n\, T_C \left(\mathcal{H}_{t_1} \dots \mathcal{H}_{t_n} \hat{O}_t \right) \qquad (20)$$

Each integral:

$$\int_c dt_i \ldots \rightarrow \underbrace{\int_{t_0}^{t} dt_i \ldots}_{\text{upper branch}} + \underbrace{\int_{t}^{t_0} dt_i \ldots}_{\text{lower branch}} = \int_{up} dt_i + \int_{lo} dt_i \qquad (21)$$

↳

$$\hat{T}_c(\ldots) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{i}{\hbar}\right)^n \left(\int_{up} dt_1 + \int_{lo} dt_1\right)\left(\int_{up} dt_2 + \int_{lo} dt_2\right)\ldots \times$$

$$\times \hat{T}_c\left(\mathcal{H}_{t_1}\mathcal{H}_{t_2}\ldots\mathcal{H}_{t_n}\right)\hat{O}_t$$

Each n-~~term~~th contribution contains all sorts of up/lo terms in which times are differently ordered: the time goes from left to right on the ~~up~~ "up" and in the reverse direction on the "lo". Times on "lo" are after (later) times at "up".

2^n-terms in the sum for each given $n = 0, 1, \ldots$
Consider one term containing m "ups" and $(n-m)$ "los":

$$\int_{up} dt_1 \ldots \int_{up} dt_m \int_{lo} dt'_1 \ldots \int_{lo} dt'_{n-m} \underbrace{\tilde{T}\left(\mathcal{H}^{\dagger}_{t'_1}\ldots\mathcal{H}^{\dagger}_{t'_{n-m}}\right)}_{\text{highest times}} \hat{O}_t \underbrace{\hat{T}\left(\mathcal{H}_{t_1}\ldots\mathcal{H}_{t_m}\right)}_{\text{lowest times}}$$

$$= \Big[\int_{lo} dt'_1 \ldots \int_{lo} dt'_{n-m} \underset{\text{antichonological}}{\tilde{T}}\left(\mathcal{H}^{\dagger}_{t'_1}\ldots\mathcal{H}^{\dagger}_{t'_{n-m}}\right)\Big] \hat{O}_t \Big[\int_{up} dt_1 \ldots \int_{up} dt_m \underset{\text{chronological}}{\hat{T}}\left(\mathcal{H}_{t_1}\ldots\mathcal{H}_{t_m}\right)\Big]$$

There are C^n_m terms like this giving the same contribution (diff. notations for times, but these are dump integration variables!)

↳ $C^n_m = \dfrac{n!}{m!(n-m)!}$

~~Also~~, Then, the sum over n can be written as:

$$\sum_{n=0}^{\infty}\sum_{m=0}^{n}\underbrace{\binom{n}{m}}_{C^n_m}\underbrace{f_{m,n-m}}_{\text{all the integrals}} = \sum_{m=0}^{\infty}\sum_{k=0}^{\infty}\binom{k+m}{m} f_{m,k} \qquad n = k+m$$

$$\hookrightarrow \sum_{n=0}^{\infty}\frac{1}{\underset{\boxed{(k+m)!}}{n!}}\left(-\frac{i}{\hbar}\right)^n \cdots = \sum_{m=0}^{\infty}\sum_{k=0}^{\infty}\underbrace{\frac{(k+m)!}{k!\,m!}}_{\binom{k+m}{m}}\frac{1}{(k+m)!}\times\left(-\frac{i}{\hbar}\right)^{k+m}$$

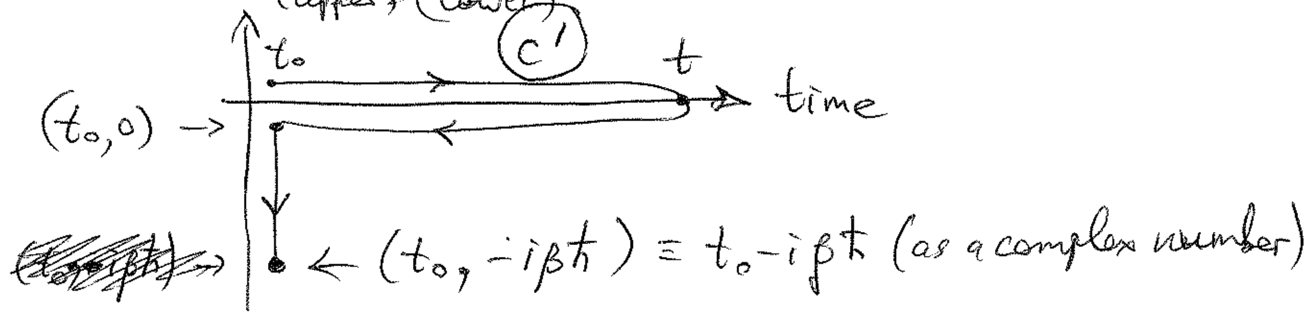
$$\times\ \tilde{T}\left(\int_{lo} dt_1' \ldots \int_{lo} dt_k'\, \mathcal{H}_{t_1'}\ldots\mathcal{H}_{t_k'}\right)\hat{O}_t\left(\hat{T}\int_{up} dt_1 \ldots \int_{up} dt_m\, \mathcal{H}_{t_1}\ldots\mathcal{H}_{t_m}\right)$$

$$= \left[\sum_{k=0}^{\infty}\frac{1}{k!}\left(-\frac{i}{\hbar}\right)^k \tilde{T}\int_{lo} dt_1' \ldots \int_{lo} dt_k'\, \mathcal{H}_{t_1'}\ldots\mathcal{H}_{t_k'}\right]\hat{O}_t\left[\sum_{m=0}^{\infty}\frac{1}{m!}\left(-\frac{i}{\hbar}\right)^m \hat{T}\int_{up} dt_1 \ldots \mathcal{H}_{t_1}\ldots\right]$$

$$= \underbrace{\left(\tilde{T}\exp\left[-\frac{i}{\hbar}\underbrace{\int_{lo}}_{\text{goes from } t \to t_0}\mathcal{H}_\tau\, d\tau\right]\right)}_{U(t_0,t)}\hat{O}_t\underbrace{\left(\hat{T}\exp\left[-\frac{i}{\hbar}\underbrace{\int_{up}}_{t_0 \to t}\mathcal{H}_\tau\, d\tau\right]\right)}_{U(t,t_0)}$$

$$\equiv U(t_0,t)\,\hat{O}_t\,U(t,t_0), \qquad \blacktriangleleft \quad (\text{Q.E.D.})$$

- In (17) the U_0-bit can also be included to other bits as the full contour: ~~$t_0 \to t \to t_0 \to t_0 - i\beta\hbar$ (upper) (lower)~~



Finally,

$$\boxed{\langle \hat{O} \rangle_t = \frac{\mathrm{Tr}\left\{ T_{C'}\left[\exp\left(-\frac{i}{\hbar}\int_{C'} \mathcal{H}_\tau \, d\tau\right) \hat{O}_t \right] \right\}}{\mathrm{Tr}\left\{ T_{C'} \exp\left(-\frac{i}{\hbar}\int_{C'} \mathcal{H}_\tau \, d\tau\right) \right\}}} \qquad (22)$$

Here $\mathcal{H}_\tau$ on the vertical (complex) part $\to \mathcal{H}_{t_0}$. Also, in the denominator,

$$T_{C'} \exp\left(-\frac{i}{\hbar}\int_{C'} \mathcal{H}_\tau \, d\tau\right) \Leftarrow U_0(t_0 - i\beta\hbar, t_0) \underbrace{U(t_0, t)\, U(t, t_0)}_{=\underline{1}} \qquad (23)$$